Technical Note

CFD Study of Merging Turbulent Plane Jets

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Abstract: A series of two-dimensional computational fluid dynamics (CFD) simulations of merging turbulent plane jets were conducted to establish the point at which the jets can be regarded as having merged to form a single jet. Simulations were conducted to quantify the influence of the nozzle width, jet axis separation, Reynolds number, and the relative jet strength on the distance from the source to the merger point. The simulation results were then compared with experimental results published in literature. As the jet fluid exited the two nozzles, the shear layer between the jet and the ambient fluid drove the ambient fluid entrainment into the jets. However, a finite volume of fluid trapped was between the two jets, and a recirculation zone with two counterrotating vortices was established. This vortex pair reduced the pressure between the jets, drawing them together more rapidly than theoretical models predicted. The simulation results indicated that equal-strength jets merged at a distance downstream of 1.4 times the initial nozzle centerline separation provided that the nozzle-separation-to-nozzle-width ratio was greater than 18. As the source momentum flux of one jet decreased relative to the other, the relative distance to merger also decreased. However, for source momentum flux ratios between 0.2 and 0.9, the merger distance was only a very weak function of the momentum flux ratio. The CFD-measured merging distance was significantly closer to the jet nozzles than current models predict. Therefore, the mixing of pollutants by using parallel plane jets is less efficient than currently thought. **DOI: 10.1061/(ASCE)HY.1943-7900.0000308.** *© 2011 American Society of Civil Engineers.*

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Introduction

Turbulent jets play a significant role in both industrial and environmental flows in a broad range of applications from the outlet of air-conditioning vents (Sandberg et al. 1992) to volcanic eruptions (Woods 1988). Turbulent jets are also regarded as a canonical turbulent shear flow and have been the subject of much research (List 1982). Jet interaction is also a common occurrence, particularly in industrial contexts in which multiple jets form in close proximity such as lines of diffusers at wastewater treatment plants (Davidson et al. 1994) and at lines of fuel nozzles in gas-fired furnaces (Szegö et al. 2009).

Despite these practical applications, there has been minimal research on how turbulent jets interact with one another. The interaction of two round turbulent jets has been studied experimentally (Yin et al. 2007; Rathakrishnan et al. 1989). Yin et al. (2007) found that the jets merged closer to the jet nozzle as the nozzles were brought closer together, though few quantitative results were presented. Rathakrishnan et al. (1989) also examined round twin jets and concluded that the merging distance is independent of the nozzle separation; however, the low resolution and narrow parameter range and the extremely low spatial resolution of their measurements calls in to question their conclusions.

Merging plane jets were examined experimentally by Miller and Comings (1960) and others (Anderson and Spall 2001; Murai et al. 1976; Lin and Sheu 1990; Nasr and Lai 1997; Fujisawa et al. 2004). These studies measured the velocity and pressure in the flow field downstream of two long thin nozzles focusing exclusively on equally turbulent plane jets and typically covering only a small number of slot widths and nozzle separation distances. For example, Miller and Comings (1960) conducted a single experiment at a Reynolds number of 17,800 by using two 1.3 cm wide nozzles with a centerline-to-centerline separation of 7.6 cm. These experimental studies have been complemented by computational fluid dynamics (CFD) studies (Anderson and Spall 2001; Lai and Nasr 1998) that again focused only on equal jets and for only a small number of nozzle-width-to-centerline separation ratios.

Of all the papers listed previously, no comprehensive study of the merging of equal and unequal turbulent jets exists that fully examines the role of the nozzle width, nozzle-centerline separation, or the ratio of the two jets source momentum fluxes. It is the goal of this study to use CFD simulations to fully explore these parameters. This paper builds on the previously published studies (Miller and Comings 1960; Anderson and Spall 2001; Murai et al. 1976; Lin and Sheu 1990; Nasr and Lai 1997) and compares their results with the simulations performed for this study. This study was conducted by using the *ANSYS FLUENT* software package (http://www.ansys.com/products/fluid-dynamics/fluent/).

Problem Definition

This study considers the flow field formed by two parallel plane jets issuing from slot nozzles of infinite length set into a vertical wall. The jets enter into a stationary ambient of the same fluid. The density of all fluids is identical so that no gravitational effects are present. The two jets have slot widths D_1 and D_2 and source velocities u_1 and u_2, respectively. The jet inlet velocity is assumed

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to be uniform across the width of the nozzle. The jet axes are separated by a distance Z. The jets are assumed to be fully turbulent so that viscous forces can be ignored. Fig. 1 represents the problem considered.

As the jet fluid moves away from the nozzles, a shear layer forms and ambient fluid is entrained into the jets; the centerline velocity decreases and a bell-shaped velocity profile develops. The finite volume of fluid trapped between the two jets causes a recirculation zone, and the jets are drawn toward each other. Further downstream, the jet-velocity profiles merge such that only a single peak in the velocity profile exists. At this stage, the jets can be considered merged and the flow behaves as a single jet. The goal of this study is to establish the distance (X_m) from jet nozzles to the merging point.

The distance (X_m) from the nozzle to the point where the velocity profiles merge (i.e., where only a single velocity peak is) is a function of each jet's source conditions and the distance between the jets' axes. That is

$$X_m = f(D_1, M_1, D_2, M_2, \mu, \rho, Z) \tag{1}$$

where $M_i = u_i^2 D_i$ denotes the specific momentum flux per unit width at the source. Dimensional analysis allows Eq. (1) to be reduced to

$$\chi_m = \frac{X_m}{Z} = f(\zeta, \psi, \mathsf{R}_1, \mathsf{R}_2) \tag{2}$$

where R = Reynolds number; $\psi = M_2/M_1$ is the ratio of the source momentum fluxes; and $\zeta = z/\sqrt{D_1/D_2}$ is the scaled nozzle separation. Without a loss of generality, we assume that Jet 1 has the higher momentum flux so that $\psi < 1$.

As stated earlier, we assume that the Reynolds number of each jet is sufficiently high such that viscous forces can be ignored and the flow is Reynolds number independent. Further, provided that the separation between the nozzles is considerably larger than the nozzle widths, the jets can be assumed to be issuing from point sources and the merging distance is independent of ζ. Therefore, for large nozzle separations, the merging distance is given by

$$\chi_m = f(\psi) \tag{3}$$

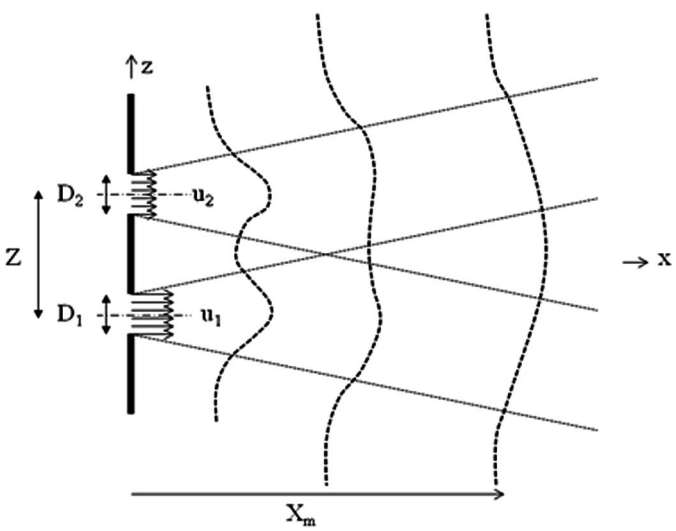

Fig. 1. Problem definition showing two unequal slot jets; dashed lines are approximate velocity profiles and the distance to merger where the velocity profile has a single peak is shown as X_m

The only theoretical model for such jet merger is that attributable to Reichardt (Maxwell 1970; Wang and Davidson 2003). This model demonstrates that the addition of the local specific momentum flux profiles of the individual jets gives the specific momentum flux profile of the combined jet flow. However, the model assumes that the jets interact through the superposition of their specific momentum flux profiles but that no dynamic interaction is between the jets attributable to their entrainment velocities. Clearly, for the case of two-dimensional (2D) plane jets, the assumption that there is no dynamic jet interaction will not hold because of the trapped fluid between the jets and the low pressure circulation zone near the nozzles. The Reichardt prediction is presented because it provides a useful introduction to the definition of the merging distance and further allows a comparison between the 2D and the three-dimensional (3D) cases in which the Reichardt model is more appropriate.

Approximating the jet velocity profile as a Gaussian curve, the velocity at any point, for two jets with negligible nozzle width, is given by

$$u(x,z)^2 = u_{c1}^2 \{ e^{-[2(z-Z/2)^2]/b(x)^2} + \psi e^{-[2(z+Z/2)^2]/b(x)^2} \} \tag{4}$$

where $b(x)$ = local jet width, z = transverse coordinate and u_{c1} = local centerline velocity jet one would have in the absence of Jet 2. For a fully developed turbulent plane jet, the local jet width is a linear function of the distance from the source and is given by (Lee and Chu 2003)

$$b(x) = 0.12x \tag{5}$$

The merging distance is defined as the distance in the x-direction in which the velocity profile has only a single peak. This is the point at which the trough between the two peaks on the velocity profile disappears. Therefore, their axial velocity gradient in the transverse direction is zero (i.e., local maxima or minima) and the curvature is zero (i.e., transition from minima to maxima). That is

$$\frac{\partial u}{\partial z} = \frac{\partial^z u}{\partial z^2} = 0 \tag{6}$$

This is the same definition suggested in a study of round turbulent plume merger (Kaye and Linden 2004), though in that case, the buoyancy profile, rather than the specific momentum profile, was used. In Kaye and Linden (2004), the buoyancy profile, as opposed to the velocity profile, was chosen because the plume flow is driven by its buoyancy. By using the same reasoning, the specific momentum profile is the appropriate profile to measure for jets because they are driven by momentum, though we note that other writers (Anderson and Spall 2001) define the merging point as coincident with the stagnation point.

For equal jets, that is $\psi = 1$, it is possible to establish an analytic solution to Eq. (6) given Eq. (5). However, for unequal jets, this is not possible because the symmetry of the problem is broken. Instead, analytic expressions were established for du/dz and d^2u/dz^2 and these were evaluated by using a grid with 500 points between the jet centerlines and twice this resolution in the streamwise direction. A numerical search was then conducted by using the grid until the points that satisfy Eq. (6) were found for 1,001 different values of ψ. The results of this search are shown in Fig. 5 along with the CFD simulation results.

CFD Simulations

To model the merging of two turbulent plane jets as described previously, the CFD software package *ANSYS FLUENT* version 6.3.26 was used. The flow was modeled as a two-phase flow, i.e., water and

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air on top of the water. For the initial condition, the virtual tank was filled with water to 1.5 m above the tank's bed and the remainder of the domain was filled with air. Hydrostatic pressure was applied to the entire domain as the initial pressure condition and the volume of fluid method was used to model the free surface between air and water. The standard $k-\varepsilon$ model was used for modeling the turbulence in the domain. This is consistent with previous studies of merging turbulent plane jets that found $k-\varepsilon$ models to produce realistic simulation results (Lai and Nasr 1998). The SIMPLE algorithm was used to model the pressure and the velocity field.

GAMBIT version 2.4.6 (http://www.fluent.com/software/gambit/index) was used to define the geometry and the mesh. Unstructured 2D triangular mesh elements were used to generate the mesh. A total of 107,386 cells were used for discretizing the equations. To test for grid independence, the $\zeta = 6$ case was repeated by using a finer mesh with 386,126 cells. The finer mesh gave the same results as the coarser mesh so the remaining simulations were computed with 107,386 cells.

The virtual two-dimensional domain was 2×2 m. The vertical distance between the jet axes was considerably smaller than the distance from the jets to any horizontal surface. The two jets were located on the left vertical wall of the tank, $1 \pm Z/2$ m above the base of the tank where Z = distance between the centerlines of the two jets. Unlike round turbulent jets, the entrainment field does not decay with distance from the jet center line. Therefore, ensuring that the entrainment flow is normal to the jet axis can be problematic. The introduction of the jets into any size virtual domain will result in a large-scale circulation. Provided the domain is large enough, the large-scale circulation will have a much larger scale than the jet axis separation. Our simulations indicated that this produces the desired entrainment flow normal to the jet axis. [See Fig. 2(a).]

To validate the CFD model, our results were compared to the previously published experimental and computational results (Miller and Comings 1960; Anderson and Spall 2001; Murai et al. 1976; Lin and Sheu 1990; Nasr and Lai 1997). For all simulations, the jet entrance Reynolds number was $\mathsf{R} = 17{,}400$ as used by Miller and Comings (1960) except for two simulations performed

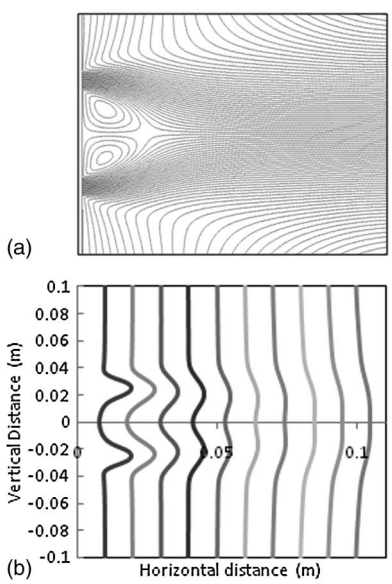

Fig. 2. (a) Streamlines for the flow attributable to two equal slot jets showing the recirculation zone between the two jets; (b) velocity profiles for two equal turbulent slot jets at a different distance downstream of the nozzles

to examine the potential role of the Reynolds number on the merging distance.

Equal Jet Results

A total of 27 dual jet simulations were performed for different nozzle-separation-to-nozzle-width ratio (ζ), Reynolds number (R), and momentum flux ratios (ψ). First, simulations for equal dual-plane jets were performed with different values of ζ to establish the separation distance required such that the nozzle width no longer influenced the merging distance. Second, two simulations were performed with different Reynolds numbers to verify that the merging distance was independent of R. Third, a series of simulations were performed at large ζ for a range of ψ to understand how the relative strength of the jets (i.e., their momentum flux ratio) influenced the merging distance.

Sample data for two equal plane jets is shown in Fig. 2(a). Shown in this figure are the recirculation zone that is established between the two jets and the drawing together of the two jets around this zone. A stagnation point exists at the right-hand end of the recirculation zone. From the velocity field calculated by the simulations, a series of horizontal velocity profiles, at different distances from the nozzles, were extracted to measure the distance to the point at which the jets can be considered merged, as defined in Eq. (6). A sample velocity profile is shown in Fig. 2(b). The typical bell-shaped velocity profile near the source is shown in this figure. At greater downstream distances, the velocity profiles merge into each other until only a single peak is in the velocity profile.

A series of simulations were performed in which the Reynolds number was constant and the nozzle-centerline separation and the nozzle width were varied. The goal of this series of simulations was to establish the relationship between the relative merging distance x_m and the relative nozzle separation ζ. As the relative separation increased (i.e., ζ increases), the influence of the nozzle width decreased until a limit was reached in which the merging distance was independent of ζ.

Two locations were used to identify the distance at which the jets could be regarded as having merged into a single plane jet; the distance to the stagnation point called the merge point in Anderson and Spall (2001), and the distance at which only a single peak is in the velocity profile called the combined point in Anderson and Spall (2001). These two points represent the upper and lower bounds on the "merging" region as defined in Fig. 1 of Anderson and Spall (2001). Both measures indicated that the nozzle width did not significantly influence the jet merger distance, provided $\zeta = z/\sqrt{D_1/D_2} > 18$. This is more clearly shown in Figs. 3 and 4, which show the merging distances scaled on the nozzle separation as a function of ζ. Also shown are previously published experimental results (Miller and Comings 1960; Anderson and Spall 2001; Murai et al. 1976; Lin and Sheu 1990; Nasr and Lai 1997). Clearly, good agreement exists between the CFD simulations and the previously published experimental data. Further, as expected, the relative merging distance χ_m approached a constant for a large nozzle separation ζ For relatively small separations, the relative merging distance increased with increasing nozzle separation and peaked at a relative separation of $\zeta \approx 5$. For larger ζ the merger distance decreased with decreasing nozzle width and approached a constant value of $\chi_m = 1.4$ for $\zeta > 18$.

A slightly different behavior is shown in the location of the stagnation point; see Fig. 4. The distance to the stagnation point increased monotonically and tended toward a scaled distance $\chi_s = X_s/Z = 0.75$ for $\zeta > 18$. The stagnation point results showed

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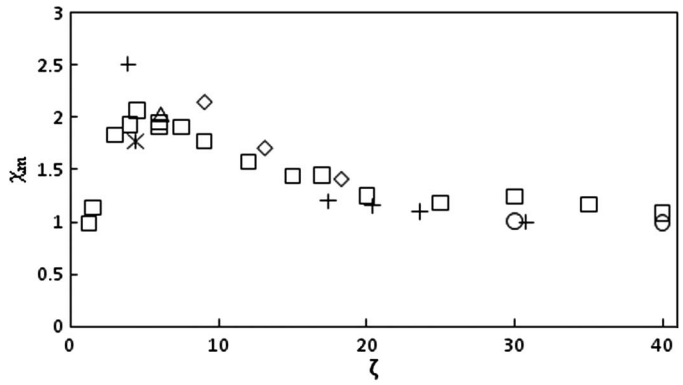

Fig. 3. Scaled merging distance χ_m as a function of the relative nozzle separation ζ; squares represent the CFD simulations from this study, and other shapes, as described subsequently, represent the previously published experimental results extracted from Fig. 7(a) of Anderson and Spall (2001); triangles (Miller and Commings 1959), diamonds (Anderson and Spall 2001), pluses (Murai et al. 1976), circles (Lin and Sheu 1990), stars (Nasr and Lai 1997)

poorer agreement with the experimental results of Miller and Comings (1960), Anderson and Spall (2001), Lin and Sheu (1990), and Nasr and Lai (1997) for smaller values of ζ though the agreement was better for larger relative nozzle separations. The poorer agreement is possibly attributable to the difficulty in accurately measuring the location of a stagnation point by using intrusive measurement probes (Khan 1989). Further, it should also be noted that far fewer experimental measurements of the stagnation point location have been reported, again likely attributable to the difficulty in measuring it.

To verify that the flow was Reynolds number independent, a series of tests were performed for a range of Reynolds numbers (i.e., $\mathsf{R} = 12{,}500$, 17,400, and 25,000) and a separation-to-nozzle-width ratio of $\zeta = 6$. The measured variation in the merging distance was less than 2% indicating that the merging distance is Reynolds number independent.

The equal dual jet study established that:

a. The simulations agree well with published experimental results,
b. The flow is Reynolds number independent for the range of Reynolds numbers used, and

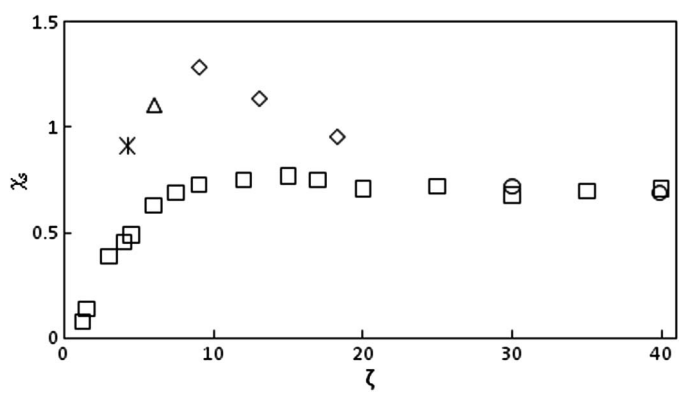

Fig. 4. Scaled distance to the stagnation point χ_s as a function of the relative nozzle separation ζ; squares represent the CFD simulations from this study, and other shapes, as described subsequently, represent the previously published experimental results extracted from Fig. 7(a) of Anderson and Spall (2001); triangles (Miller and Commings 1959), diamonds (Anderson and Spall 2001), circles (Lin and Sheu 1990), stars (Nasr and Lai 1997)

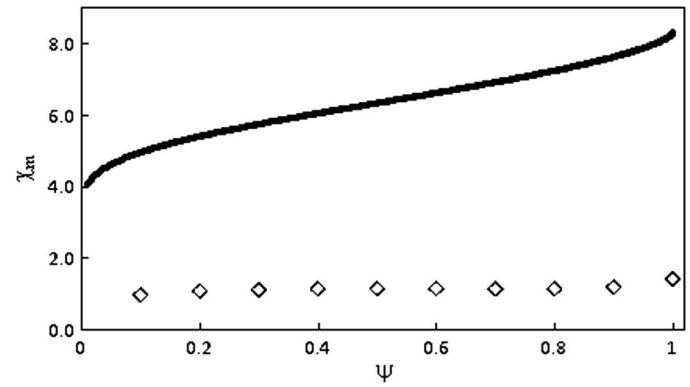

Fig. 5. Scaled merging distance χ_m as a function of the momentum flux ratio ψ; the line represents the Reichardt prediction assuming no jet interaction and the symbols represent the CFD simulation results

c. The merging distance is independent of the nozzle width provided $\zeta > 18$.

Unequal Jet Results

The goal of this portion of the study was to understand the role of the jet momentum flux ratio ψ on the merging distance when the nozzle width was small enough that it did not influence the flow, that is, for $\zeta > 18$. To that end, a series of simulations were performed in which the momentum flux ratio was varied by varying the nozzle width and the source velocity of one jet while keeping the second jet constant. Both the jet width and the source velocity were varied to adjust the momentum flux while keeping the Reynolds number of the jets constant. For all simulations, the nozzle-separation-to-nozzle-width ratio was large ($\zeta = z/\sqrt{D_1/D_2} > 18$).

The distance to the merger point as a function of the momentum flux ratio, that is $\chi_m = f(\psi)$, for large ζ, is shown in Fig. 5 along with the model prediction of Reichardt (Maxwell 1970; Wang and Davidson 2003). As expected, there was a significant disagreement between the simulation results and the Reichardt model prediction because the model assumed that the two jets did not interact and were free to grow independently of each other. This was not the case for dual plane jets for two reasons. First, each jet entrained ambient fluid from its surroundings creating an entrainment velocity field that tended to draw the jets together (Kaye and Linden 2004). Second, and more importantly, there was a finite volume of fluid trapped between the jets. The shear generated by the jets drove a recirculation flow between the jets [see Fig. 2(a)].

As the momentum flux ratio ψ was reduced from one, the merger distance dropped slightly from approximately 1.4 at $\psi = 1$ to 1.1 at $\psi = 0.9$. It then remained relatively constant as ψ was reduced further until dipping again for $\psi < 0.2$. This trend was qualitatively similar to that observed in an experimental study of merging round turbulent plumes in which the merging distance was found to be relatively constant for a broad range of buoyancy flux ratios (Kaye and Linden 2004).

For both equal and unequal plane jets, the merger point was very close to the jet source. Therefore, only a small region existed in which the ambient fluid was entrained into two jets rather than one jet. The flow rate in a single jet scaled on the square root of the momentum flux, $Q \sim M^{1/2}$. Therefore, two jets with half the momentum flux had a larger volume flux as $Q \sim 2(M/2)^{1/2} > M^{1/2}$. As the concentration of any pollutant in the jet was inversely proportional to the volume flux, two jets would result in more mixing and lower concentrations. However, as the two jets merged close to

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their source, the benefit of having two plane jets was minimized, and a single-jet model may provide a better guide to downstream pollutant concentrations.

Conclusions

A series of CFD simulations were performed to investigate the downstream distance at which two parallel turbulent plane jets could be regarded as merged. The results of the simulations showed good agreement with previously published experimental measurements of the distance to the merging point. The agreement between simulation and experiment for the location of the stagnation point was poorer for smaller ratios of the centerline separation-to-nozzle width. Both measurement criteria showed that the merging distance was independent of the nozzle width provided that the centerline separation-to-nozzle-width ratio was large (i.e., $\zeta = z/\sqrt{D_1/D_2} > 18$).

A series of unequal jet simulations was performed to examine the effect of varying the ratio of the source momentum fluxes of the two jets. The results of the simulated merging distance were compared to the theoretical prediction of Reichardt. The agreement with theory was very poor because the jets did not grow independently of each other as assumed by Reichardt, but rather were drawn together because of the low-pressure recirculation zone that formed between the jets. Further, the merging distance was found to be relatively independent of the momentum flux ratio in the range $0.2 < \psi < 0.9$.

Notation

The following symbols are used in this paper:

$b(x)$ = local jet width;
$D_{1,2}$ = slot width;
$M_{1,2}$ = specific momentum flux-per-unit-width at the source;
$\mathsf{R}_{1,2}$ = nozzle Reynolds number;
$u_{1,2}$ = source velocities;
u_{c1} = local centerline velocity Jet 1 in the absence of Jet 2;
X_m = distance from jet nozzles to the merging point;
x = distance from jet;
x_m = dimensionless distance from jet nozzles to the merging point;
x_s = dimensionless distance from jet to stagnation point;
Z = distance between jet axes;
z = transverse coordinate;
ζ = scaled nozzle separation;
μ = dynamic viscosity;
ρ = density; and
ψ = fluxes ratio.

References

Anderson, E. A., and Spall, R. A. (2001). "Experimental and numerical investigation of two-dimensional parallel jets." *J. Fluids Eng.*, 123(2), 401–406.

Davidson, M. J., Papps, D. A., and Wood, I. R. (1994). "The behaviour of merging buoyant jets." *Recent research advances in the fluid mechanics of jets and plumes*, P. A. Davies and M. J. Valente Neves, eds., Springer-Verlag, New York, 465–478.

Fujisawa, N., Nakamura, K., and Srinivas, K. (2004). "Interaction of two parallel plane jets of different velocities." *J. Visualization*, 7(2), 135–142.

Kaye, N. B., and Linden, P. F. (2004). "Coalescing axisymmetric turbulent plumes." *J. Fluid Mech.*, 502, 41–63.

Khan, A. A. (1989). "A study of intersecting circular turbulent jets." M.S. thesis, University of Alberta, Edmonton, Alberta.

Lai, J. C. S., and Nasr, A. (1998). "Two parallel plane jets: Comparison of the performance of three turbulence models." *Proc. Inst. Mech. Eng., Part G*, 212(6), 379–391.

Lee, J. H. W., and Chu, V. H. (2003). *Turbulent jets and plumes—A Lagrangian approach*. Kluwer Academic, Norwell, MA.

Lin, Y. F., and Sheu, M. J. (1990). "Investigation of two plane parallel unventilated jets." *Exp. Fluids*, 10(1), 17–22.

List, E. J. (1982). "Turbulent jets and plumes." *Annu. Rev. Fluid Mech.*, 14, 189–212.

Maxwell, W. H. C. (1970). "Flux development region in submerged jets." *J. Engrg. Mech. Div*, 96(6), 1061–1079.

Miller, D. R., and Comings, E. W. (1960). "Force-momentum fields in a dual-jet flow." *J. Fluid Mech.*, 7(02), 237–256.

Murai, K., Taga, M., and Akagawa, K. (1976). "An experimental study on confluence of two-dimensional jets." *Bull. JSME*, 19(134), 958–964.

Nasr, A., and Lai, J. C. S. (1997). "Comparison of flow characteristics in the near field of two parallel plane jets and an offset plane jet." *Phys. Fluids*, 9(10), 2919–2931.

Rathakrishnan, E., Vijayabhaskar-Reedy, P., and Pabmanaban, K. (1989). "Some studies of twin jet propagation." *Mech. Res. Commun.*, 16(5), 279–287.

Sandberg, M., Wiren, B., and Claesson, L. (1992). "Attachment of a cold plane jet to the ceiling; length of recirculation and separation distance." *Proc., ROOMVENT '92, 3rd International Conf. on Air Distribution in Rooms*, Scandinavian Federation of Heating, Ventilating and Sanitary Engineering Associations (SCANVAC), Aalborg, Denmark, 487–499.

Szegö, G. G., Dally, B. B., and Nathan, G. J. (2009). "Operational characteristics of a parallel jet MILD combustion burner system." *Combust. Flame*, 156(2), 429–438.

Wang, H. J., and Davidson, M. J. (2003). "Jet interaction in a still ambient fluid." *J. Hydraul. Eng.*, 129(5), 349–357.

Woods, A. W. (1988). "The fluid dynamics and thermodynamics of eruption columns." *Bull. Volcanol.*, 50(3), 169–193.

Yin, Z., Zhang, H., and Lin, J. (2007). "Experimental study on the flow field characteristics in the mixing region of twin jets." *J. Hydrodyn.*, 19(3), 309–313.

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